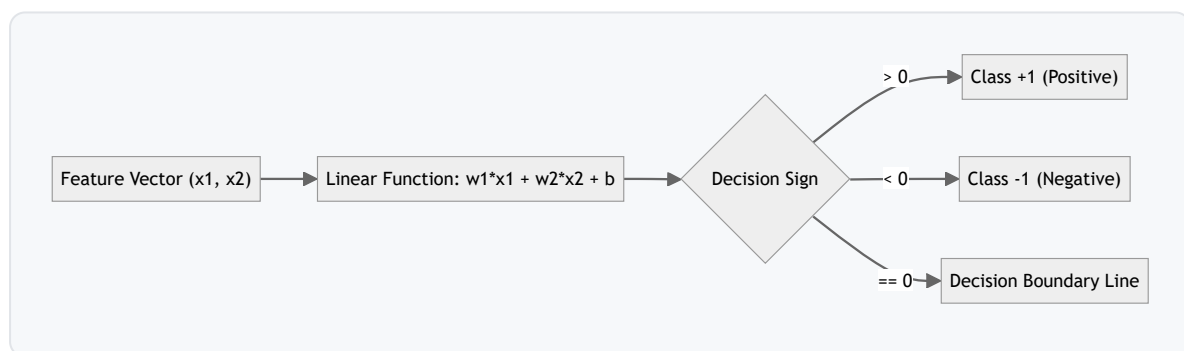


Day 21 - Lecture 21.2: Straight Lines & Linear Equations

[Python 3.10+](#)[Jupyter Notebook](#)[PDF notes.pdf](#)[Language: English | हिंदी / Hinglish | !\[\]\(e3f8612927870f2e0f9f5989e6dd3064_img.jpg\) Back to Lecture 21 Module](#)

1. Overview & Linear Decision Boundaries

In Cartesian 2D coordinate space, a **straight line** is the geometric locus of points satisfying a linear polynomial equation of degree 1. In Machine Learning, a straight line serves as the fundamental **linear decision boundary** that separates two classification classes in a 2D feature space.



2. Mathematical Formalism

1. Slope-Intercept Form:

For non-vertical lines with slope m and vertical intercept c :

$$y = mx + c$$

where the slope $m = \tan(\theta) = \frac{y_2 - y_1}{x_2 - x_1}$ measures the rate of vertical change per unit horizontal displacement.

2. General Standard Form:

$$ax + by + c = 0$$

Converting between forms:

$$y = -\frac{a}{b}x - \frac{c}{b} \implies m = -\frac{a}{b}, \quad \text{Intercept} = -\frac{c}{b} \quad (b \neq 0)$$

3. Hyperplane Representation in Machine Learning:

In machine learning vector notation, let feature vector $\mathbf{x} = [x_1, x_2]^T$, weight vector $\mathbf{w} = [w_1, w_2]^T$, and bias b :

$$\mathbf{w}^T \mathbf{x} + b = w_1 x_1 + w_2 x_2 + b = 0$$

- The weight vector $\mathbf{w}$ is strictly **perpendicular (normal)** to the straight line boundary.
 - The bias b controls the perpendicular offset of the line from the origin.
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3. Python Implementation: Plotting Linear Decision Boundary

```
import numpy as np
import matplotlib.pyplot as plt

# Linear equation parameters:  $w_1x_1 + w_2x_2 + b = 0 \rightarrow 2x_1 - 3x_2 + 6 = 0$ 
#  $w_1, w_2, b = 2.0, -3.0, 6.0$ 

# Generate  $x_1$  values
x1 = np.linspace(-5, 5, 200)
# Solve for  $x_2$ :  $x_2 = -(w_1x_1 + b) / w_2$ 
x2 = -(w1 * x1 + b) / w2

slope = -w1 / w2
intercept = -b / w2

print(f"Line Equation: {w1}*x1 + ({w2})*x2 + {b} = 0")
print(f"Slope m:      {slope:.4f}")
print(f"Intercept c:   {intercept:.4f}")
```

4. Key Takeaways & Interview Points

- **Geometric Meaning of Weights:** The weights w_1, w_2 determine the orientation (tilt) of the decision boundary, while the weight vector $\mathbf{w}$ points directly in the direction of the positive class.
- **Generalization to High Dimensions:** A straight line in $\mathbb{R}^2$ generalizes to a plane in $\mathbb{R}^3$, and a **hyperplane** in $\mathbb{R}^D$.